Lecture 39 — Least squares, data fitting, and positive definiteness

Wed Dec 2

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Reading: Boyd 12.2–12.4 and 13.1, class notes on symmetric and positive-definite matrices, Lab 5, and Lachniet Ch. 2 (least squares). Related objectives: Lectures.

Learning objectives

By the end of class, students should be able to:

1. Compute a least-squares solution with Octave or Python.
2. Plot observed and fitted values.
3. Interpret residuals and an RMS error in context.
4. Check the residual's orthogonality to the columns of A.
5. Evaluate whether a fitted model is useful rather than merely reporting coefficients.
6. Recognize that $A^\mathsf{T} A$ is symmetric positive semidefinite, explain $x^\mathsf{T} A^\mathsf{T} A x = \|Ax\|^2$, and recognize that $A^\mathsf{T} A$ is positive definite when A has linearly independent columns.